sample_1-Evaluation report

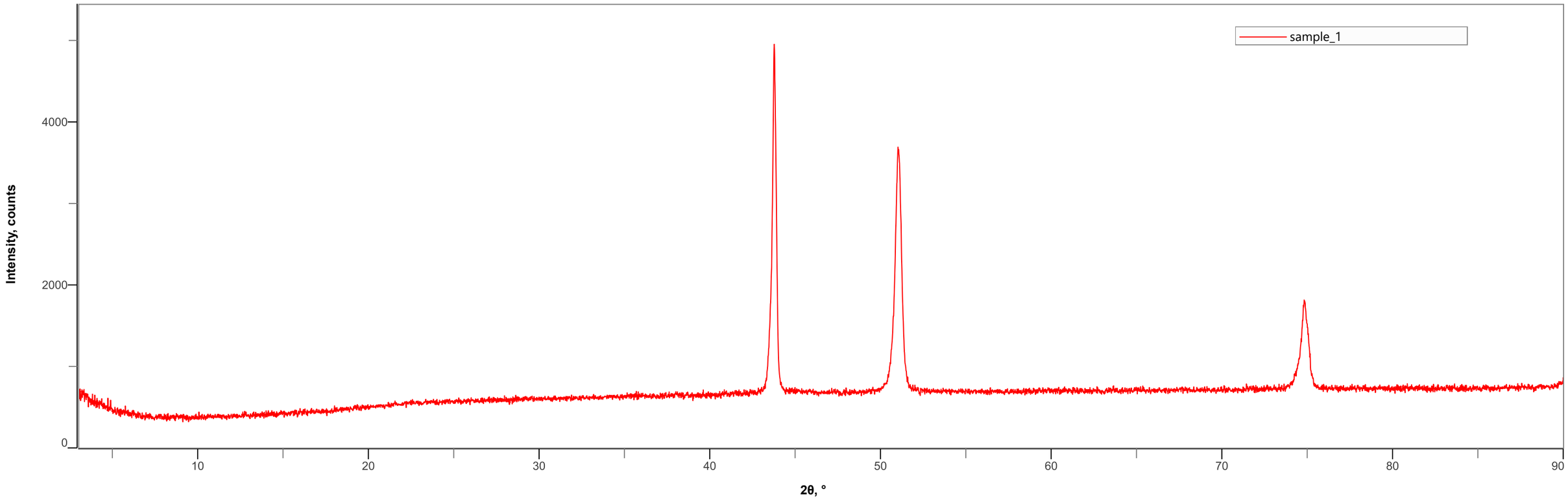
General information

Analysis date	2025-11-27 13:12:16	Measurement start time	2025-11-23 12:34:33
Analyst	Administrator	Operator	Administrator
Sample name	sample_1	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\sample_1.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	Standard sample stage	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

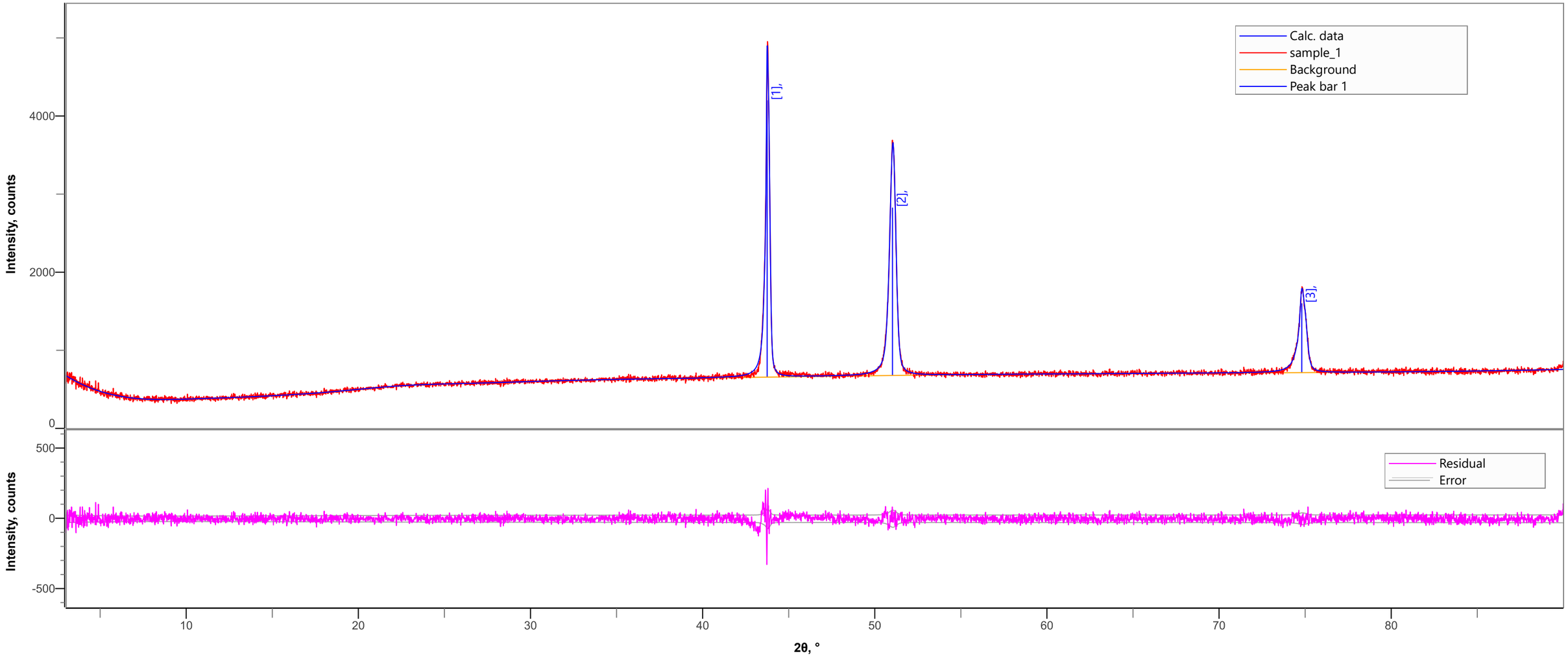
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	3.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

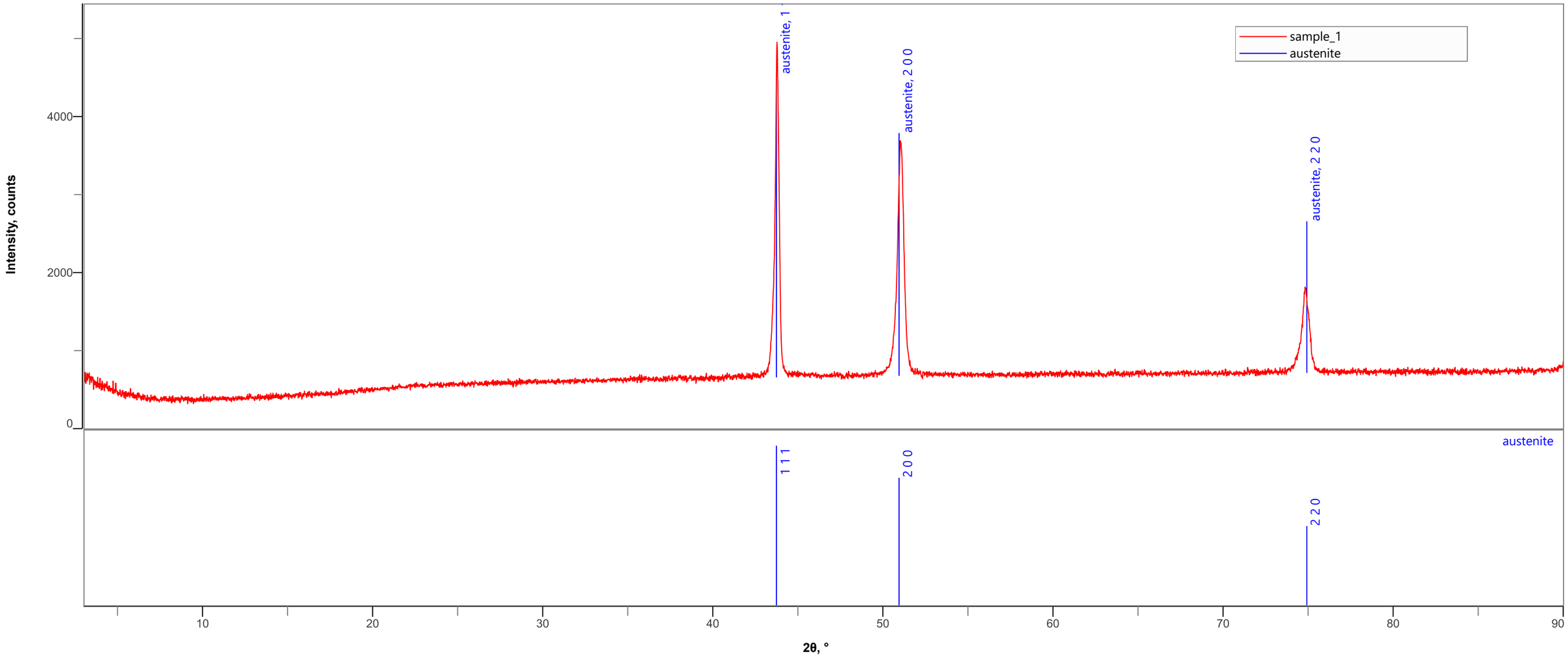
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
austenite	Cr - Ni - Fe - C	0.473	S/M:PDF-5+ 2026	225 : Fm-3m	00-031-0619

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, counts	FWHM, °	Int. I., counts°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay(ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	43.7275(11)	2.06846(5)	3538(52)	0.1764(17)	973(3)	0.275(5)	0.82(2)	1.40(2)	0.461(18)	507(5)	austenite: 1 1 1	Cr - Ni - Fe - C	00-031-0619	90.31	Split Pseudo-Voigt	-	-	-	-	-	
2	51.004(3)	1.78913(10)	2145(35)	0.384(2)	1078(4)	0.502(10)	1.32(4)	0.612(18)	0.53(2)	239.1(14)	austenite: 2 0 0	Cr - Ni - Fe - C	00-031-0619	100.00	Split Pseudo-Voigt	-	-	-	-	-	
3	74.769(5)	1.26869(7)	886(21)	0.344(8)	478(3)	0.539(17)	1.07(7)	1.25(5)	0.63(5)	303(7)	austenite: 2 2 0	Cr - Ni - Fe - C	00-031-0619	44.36	Split Pseudo-Voigt	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
austenite	3.58366	3.58366	3.58366	90.000	90.000	90.000

d-I List

austenite

No.	2θ, °	d, Å	h k l	Norm. I.
1	43.71492	2.06903	1 1 1	100.00
2	50.92162	1.79183	2 0 0	80.00
3	74.88449	1.26702	2 2 0	50.00

Simple Inverse Pole Figure

austenite

